

POWERBOX · DESIGN DECISION RECORD

PowerBox recommendations accepted

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DOCUMENT CONTROL

Document ID	PBX-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for every item that carried a recommendation; items without one remain proposed, awaiting Amish

■ Context

On 2026-09-25 Amish wrote, for every portfolio repo: "i accept all your recommendations, go with them across all repos." PBX-DDR-001 had already recorded his earlier decisions on the TRL 2 review (items 1 to 12). This record applies the new instruction to every remaining PowerBox item marked "Proposed, awaiting Amish" or "awaiting Amish" in `docs/REVIEW.md` and `docs/decisions/` that carries a recommendation. Where a recommendation offered several options, the recommended option is the decision. Items with no recommendation stay open; no choice is invented for them. TRL 4 stays on hold by Amish's instruction, and the repo stays at TRL 3.

■ Decision

Table 1. Items newly decided: "Decided by Amish, 2026-09-25: go with recommendation".

#	Item (source)	Decision	What changed in the repo
1	Recessed wake button (PBX-DDR-001 item 17, PBX-CAL-001 section 4)	A recessed, normally closed wake button in series with the 10 kΩ INTERLOCK coding resistor wakes a pack that fell asleep in the bay; no keep-alive cell, WAKE pin unused	Already in the design (<code>bom/bom.csv</code> line 11, <code>cad/src/model.py</code> output panel). Status wording updated in PBX-PRC-001 v0.5 and PBX-DDR-001 v0.2. No geometry, cost or calculation change
2	48 V to 12 V buck at 30 A (PBX-DDR-001 item 17, PBX-CAL-001 section 5)	30 A (360 W) buck for a 26.5 A worst-case demand, replacing the TRL 2 20 A unit	Already in <code>bom/bom.csv</code> line 10 (\$30) and PBX-CAL-001. Status wording updated in PBX-PRC-001 v0.5. No change to numbers
3	Fuse ratings (PBX-DDR-001 item 17, PBX-CAL-001 Table 3)	30 A main fuse and every other 48 V fuse rated 60 V DC or more with at least 1 kA breaking capacity; 20 A inverter breaker; fuse block as in PBX-CAL-001	Already in <code>bom/bom.csv</code> line 16 and PBX-CAL-001. Ratings now listed in PBX-PRC-001 v0.5 key design choices
4	TRL 2 review list, items 1 to 8, and human-powered charging (<code>docs/REVIEW.md</code> , sessions of 2026-09-24)	As recommended; already recorded in PBX-DDR-001 items 1 to 9	<code>docs/REVIEW.md</code> wording updated so the 2026-09-24 lists no longer read "Proposed, awaiting Amish" for these items

The budget is unchanged: `budget_usd` stays at \$450 for the PowerBox parts, excluding the SwapCell pack (PBX-DDR-001 item 8). No requirement text changes: R1 to R12 in PBX-REQ-001 v0.4 already assume the 30 A buck, the fuse ratings and the wake button. PBX-CAL-001, PBX-DWG-001 (Rev P1), the model and the BOM are unchanged because the accepted items were already designed in at TRL 3; only their status changes.

■ Items still open

Table 2. Items with no recommendation: "Proposed, awaiting Amish".

#	Item	Why it stays open
13	R2 shortfall (1.87 evenings against 2): relax R2, discharge to 3.5 %, or revise the reference evening with users (about 189 Wh at the loads)	No recommendation was made
14	First co-design partner	No preference stated; the portfolio picks partners per area later
15	Theft resistance (lockable pack door)	No recommendation
16	Energy metering on the display for charging points	No recommendation
18	SwapCell clarification: may a pack in legacy discharge (state 5) accept a station heartbeat and move to mode 2 or 4 without opening its output?	A question for SwapCell, not a recommendation; listed as a cross-repo action in docs/REVIEW.md

■ Consequences

- R2 stays not met, and R6 and R11 stay at risk, because none of the accepted items changes their numbers.
- Work that the accepted items imply beyond paper (buying the fuses, buck and button, wiring the INTERLOCK loop, bench tests) is TRL 4 and is on hold by Amish's instruction.